Name: ____________________

MULTIPLICATION AND DIVISION, (DIVISION AS THE INVERSE)

Date: ________________

LO: I can use multiplication facts to solve division problems by recognising they are inverse operations.

What does inverse mean?

Inverse means **opposite**. Multiplication and division are inverses: one undoes the other.

Fact families: from one multiplication you can write FOUR related facts. e.g. 3×4=12 gives 4×3=12, 12÷3=4, 12÷4=3.

Fact families

🟢	5 × 6 = 30	multiplication
🟢	6 × 5 = 30	commutative
🟢	30 ÷ 5 = 6	inverse
🟢	30 ÷ 6 = 5	inverse
🔴	30 ÷ 30 = 1 × 5 + 6	doesn't follow

Key idea

If a × b = c, then c ÷ a = b AND c ÷ b = a. Use what you know about times tables to divide.

7 × 8 = 56 **56 ÷ 7 = 8**

Same numbers, inverse operation

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – fact family from a multiplication

If 4 × 7 = 28, write the four related facts.

4 × 7 = 28

7 × 4 = 28

28 ÷ 4 = 7, 28 ÷ 7 = 4

Four facts

Same three numbers (4, 7, 28) just rearranged.

Example 2 – use multiplication to divide

Calculate 42 ÷ 6

Think: 6 × ___ = 42

From times table: 6 × 7 = 42

So 42 ÷ 6 = 7

= 7

Recall the multiplication fact, then read off the inverse.

Example 3 – checking with the inverse

Check that 81 ÷ 9 = 9 is correct.

Inverse: 9 × 9 = ?

9 × 9 = 81

Match

Correct

If multiplication gives the original, the division is right.

Example 4 – missing-number problem

Solve: ___ ÷ 8 = 7

Inverse: 8 × 7 = ?

8 × 7 = 56

So the missing number is 56

= 56

Multiplying both sides by 8 gives the original number.

YOUR TURN – PRACTICE (deliberate practice)

1. Complete the fact family
Each multiplication has 3 partners. Write them.

a) 5 × 9 = 45 ___, ___, ___

b) 6 × 8 = 48 ___, ___, ___

c) 7 × 7 = 49 ___, ___

d) 12 × 4 = 48 ___, ___, ___

2. Use multiplication to divide
Think which multiplication helps.

a) 36 ÷ 4 = ___

b) 63 ÷ 7 = ___

c) 72 ÷ 8 = ___

d) 55 ÷ 5 = ___

3. Check with the inverse
Is each statement correct? Use multiplication to check.

a) 84 ÷ 7 = 12 — check by 7×12

b) 54 ÷ 6 = 8 — check by 6×8

c) 121 ÷ 11 = 11 — check

d) 96 ÷ 8 = 11 — check

4. Missing-number problems
Use the inverse to find the unknown.

a) ___ ÷ 9 = 8

b) 144 ÷ ___ = 12

c) ___ ÷ 7 = 11

d) 108 ÷ ___ = 9

5. Mixed fact-family practice
Spot the multiplication fact you need.

a) If 12 × 6 = 72, then 72 ÷ 6 = ___

b) If 8 × 9 = 72, then 72 ÷ 9 = ___

c) 11 × 11 = ___, so 121 ÷ 11 = ___

d) If 7 × 5 = 35, then 35 ÷ 5 = ___

e) Write 4 facts from 4 × 6 = 24

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Division is the inverse of multiplication ☐

Correct answer: __________

Reason: __________________

b) If 7×9=63, then 63÷7=9 ☐

Correct answer: __________

Reason: __________________

c) Division is commutative: 12÷3 = 3÷12 ☐

Correct answer: __________

Reason: __________________

d) If a×b = c, then c÷a = b ☐

Correct answer: __________

Reason: __________________

e) Multiplication undoes division ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A teacher has **56 stickers** to share equally between **8 children**. Without dividing directly, use multiplication facts to find how many each child gets. Show your reasoning.

Expression: ______________________________

Simplified: ______________________________

b) Mia says 'I know 7 × 11 = 77, so I also know 77 ÷ 7 and 77 ÷ 11.' Is she right? Write all four related facts and explain.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!